A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 23

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 7×10^{11} tonnes **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 10% of 10 000 = 1000 billion tonnes **1** 1000 / 700 x 100 = 143% (accept 142.9) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) **1** Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** CO2 dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis **1** Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons **1** When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.5	**1 mark each** 1 mark for each of the 2 points below. **1** Aerobic respiration uses oxygen to break down organic compounds completely to CO_2 (and water), releasing much more energy **1** Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH_4), which is later oxidised to CO_2 in the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** The readings are precise but may not be accurate **Also accept** • **precise** → readings close together / fine resolution • **accurate** → close to the true value	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** A result that differs from other results with which it was expected to be similar **1 Repeat the study / measurement:** if it recurs it is a 'real' result caused by another variable – if not it was an error and can be excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** To test the reliability of the method / equipment **1** To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** They store chemical energy that is easy to keep until it is needed, and easy to convert into the heat that is usually wanted **1** They have a high energy density, so they reach the high temperatures needed to smelt metal ores and to raise high-pressure steam for turbines **1** High energy density also makes them efficient to transport, because a small mass or volume carries a lot of energy – 75 t of aviation fuel can carry a 400 t aircraft 5600 km **1** They were often found in very abundant local deposits, so industry could grow up on the coalfield that supplied it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 75 t / 400 passengers = 0.1875 t per passenger **1** = 188 kg (accept 187 - 190 kg) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (kg) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
03.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Many deposits are too deep, too small or in places that are too difficult to reach for the technology available **1** A lot of natural gas is held in fine-grained impermeable shale, from which it cannot flow to a conventional well **1** Oil shale holds more than the world's total reserves of crude oil, but the hydrocarbons are solid and insoluble, so the rock has to be heated to melt the oil before it will flow **1** Reserves count only what can be extracted economically, so deposits that would cost more than the fuel is worth are excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	3

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Economic decisions are made on the cheapest option available at the time rather than on long-term supply, so the mines closed while coal was still in the ground **1** The abandoned mines have flooded and are unsafe, so the remaining coal cannot be reached even though North Sea gas will be exhausted within the next thirty or forty years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.5	**Any 5 of 7** 1 mark each for any 5 of the 7 points below, in any order, maximum 5. **1** Deep mining damages the surface through loss of habitat, the transport infrastructure it needs and subsidence above the worked seams **1** Opencast mining causes greater habitat damage because all the overburden above the coal has to be cleared **1** Spoil heaps from mining occupy land, can be unstable and are slow to revegetate **1** Oil spills from rigs pollute the sea, and oil-based drilling mud used to lubricate the drill pipes can pollute groundwater, rivers and the sea **1** Surplus gas on oil rigs is flared to reduce the risk of explosion, releasing sulfur dioxide, carbon dioxide and smoke **1** Pipeline construction damages habitat over a long, wide corridor, although careful route planning, soil replacement and replanting reduce this **1** Fracking to release shale gas uses large volumes of water, risks contaminating groundwater and can cause local earth tremors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Fracking** → pumping fluid into rock at high pressure to release gas or oil	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-190; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 4 points below. **1 Production prevention:** desulfurisation of fuels before combustion – low-temperature (fluidised bed) combustion to prevent NOx – smokeless fuels **1 Prevention of release:** electrostatic precipitators / bag filters / cyclones for smoke – catalytic converters – flue gas desulfurisation – bund walls – landfill liners **1 Post-release remediation:** oil spill clean-up (booms, skimmers, dispersants, bioremediation) – phytoremediation of metal-contaminated land – liming of acidified lakes **1 Alternative processes:** electric vehicles instead of petrol / diesel – renewable energy instead of fossil fuels – pyrethroids instead of organochlorines – fuel cells instead of internal combustion engines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	4
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Remediation is often incomplete / cannot recover all the pollutant (eg oil spills, dispersed pollutants), so some damage still occurs **1** Prevention avoids the damage entirely and is usually cheaper than clean-up / avoids polluter-pays costs – moving from post-production treatment to non-release is a circular economy principle **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Satellites detect and track the extent / movement of oil slicks over large areas, quickly and repeatedly, so clean-up can be directed and polluters identified **1** GPS tracking monitors ship positions so navigation errors / collisions / groundings are reduced and offshore shipping routes are enforced **1** Polymers with a high surface area adsorb metal ions from contaminated water / effluent so they can be removed and recovered **1** Plants that absorb heavy metals (hyperaccumulators) are grown on contaminated land, harvested and disposed of / metals recovered, cleaning the soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) **1** Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (200 - 25) / 200 x 100 **1** = 87.5% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) **1** Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths **1** Higher doses caused sterility / death **1** After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) **1** Breeding success recovered so the population rose **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	4
05.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** DDT persists in soil / sediments / food chains for many years so exposure continued **1 Birds of prey are k-selected:** few young per year, slow to mature so populations grow slowly **1** Other pressures (habitat loss, persecution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** The void fills with water naturally – wetlands are uncommon / valuable habitats in most regions **1** The new habitat may have greater wildlife value than the farmland lost – low cost compared with filling and levelling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1 Housing:** ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) **1 Farmland:** landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	1 mark for the point below. **1** Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost – transplanted plants often fail **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
07.1	**1 mark each** 1 mark for each of the 3 points below. **1** UV light splits diatomic oxygen (O2) into two monatomic oxygen atoms (O) **1** Monatomic oxygen reacts with diatomic oxygen to form ozone (O3) **1** UV light also splits ozone into O2 and O – the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.2	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Useful:** boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants) – dissolve grease but do not damage electrical components (solvents) – non-flammable – non-toxic **up to 2 Persistence:** chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere **up to 2 Dissociation by UV:** in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals **up to 2** Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation – chlorine is regenerated so one atom destroys many ozone molecules – bromine / iodine act similarly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **Persistence** → does not break down / stays in the environment for a long time **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) **1** Damage to leaf tissue and reduced photosynthesis / crop yields (plants) **1** Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.4	1 mark for the point below. **1** An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
07.5	1 mark for the point below. **1** Concentration varies with altitude and the stratosphere is not normally sampled – Dobson Units measure the total column thickness instead **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
07.6	1 mark for the point below. **1** The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - tiny pores between clay particles allow capillary rise – none in sand	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 3 points below. **1 Hygroscopic:** held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants **1 Capillary:** held in pores by surface tension – does not drain but can be absorbed by plants **1 Gravitational:** drains freely through large pores – available to plants only briefly as it passes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.3	1 mark for the point below. **1** Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **peds** → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** The climate naturally fluctuates (solar output, Earth's orbit) so extreme events can occur without human influence **1** An individual event, or even several, does not show a trend – the change may be in the frequency of events which can only be assessed years later **1** Changes occur at different time and spatial scales (local storm vs global trend) – interconnected systems make attribution complex **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity	3
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Towns have a heat-island effect (heat emissions, dark surfaces absorbing light) **1** As the town grows around the weather station the warming recorded is partly local, not global – older equipment / sparse global coverage also reduce reliability **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Marks shared** Up to 5 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Tree rings:** width of each annual ring shows the growth rate, indicating temperature / rainfall in that year – rings easily dated by counting – limitation: growth also affected by other factors (water, pests, shading), only covers the life of the tree / limited regions **up to 2 Ice cores:** annual snow layers compact into ice – air bubbles trapped in the ice give the CO2 (and methane) concentration when trapped – oxygen isotope ratios indicate temperature – up to 800 000 years (Dome C) – limitation: deep layers compressed / less distinct, dating needs radio-isotope analysis and volcanic ash layers – only polar regions **up to 2 Others accepted:** coral growth rings – pollen in lake sediments dated by radiocarbon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 100-102; AQA spec 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Uses / benefits:** nuclear electricity (low carbon, high energy density), medicine (X-rays, CT, radiotherapy, tracers, sterilising equipment), industry (thickness gauges, polymer cross-linking, testing turbines, oil exploration), agriculture (food irradiation, sterile male technique, mutation breeding), research tracers, smoke detectors, ship / submarine propulsion • **Risks:** free radical production, DNA damage, mutation, cancer – acute and chronic, somatic and gonadic effects – contamination vs exposure – activation products – long-lived wastes – accidents (Chernobyl, Fukushima) – weapons fallout • **Factors affecting risk:** type of radiation (alpha, beta, gamma, neutrons - penetration, RBE / weighting factors), half-life, dose (Gy, Sv), route of exposure • Risk:benefit analysis and why it is difficult (latency, alternative causes, poor exposure data, risks and benefits to different people, offsets such as avoided surgery) • **Principles:** ALARA, BATNEEC – closed sources, remote handling, absorbers / shielding, distance (inverse square law), time, protective clothing, decontamination, potassium iodide • **Monitoring:** dosemeters, film badges, air and contamination monitors – environmental monitoring via CPA and CGM – waste categories and disposal (vitrification, encapsulation, Drigg) • **Evaluation / judgement:** benefits generally outweigh risks when controls are applied – public perception vs actual risk (background radon 50% of dose vs <1% from nuclear industry)	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **half-life** → the time for the activity, or the amount, to fall by half • **vitrification** → sealing waste into glass **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	
	Model answer - Level 3 (8 marks)	
	Ionising radiation is used because it does things nothing else can: nuclear fission generates low-carbon electricity from fuel of extraordinary energy density; medicine relies on X-rays and CT scans, radiotherapy, tracers and gamma sterilisation of equipment; industry uses thickness gauges, cross-links polymers, tests turbine blades and explores for oil; and agriculture uses irradiation to preserve food and sterilise insect pests. The risks arise because radiation produces free radicals that damage DNA, causing mutation and cancer, with acute and chronic, somatic and gonadic effects; contamination is worse than exposure because the source stays with the person; and activation products, long-lived wastes, accidents at Chernobyl and Fukushima and weapons fallout add environmental risks.	
	Risk depends on the type of radiation - alpha is barely penetrating but has a weighting factor of 20 - on half-life, on dose in grays and sieverts and on the route of exposure. Weighing risks against benefits is difficult because cancers appear years later with many other possible causes, exposure data are poor, the people who benefit are often not those at risk, and benefits such as avoided surgery are hard to quantify.	
	Control follows ALARA and BATNEEC: closed sources, remote handling, shielding, distance under the inverse-square law, limiting time, protective clothing, decontamination and potassium iodide against iodine-131. Workers wear dosemeters and film badges and pass contamination monitors; discharges are managed by Critical Pathway Analysis and Critical Group Monitoring; wastes are vitrified, encapsulated or stored at licensed sites. With these controls the benefits generally outweigh the risks - natural radon supplies about half the average dose and the nuclear industry under one per cent - although public perception of risk remains far higher than the measured risk.	
	Weaker answer - Level 2 (5 marks)	
	Ionising radiation is used in nuclear power stations to make electricity, in hospitals for X-rays and treating cancer, and in industry. The risks are that radiation damages DNA and causes cancer, and accidents like Chernobyl release radiation into the environment. Nuclear waste stays dangerous for a long time. Workers are protected by shielding, keeping their distance and wearing film badges to measure their dose. The public is protected by monitoring around nuclear sites. Overall the benefits are worth the risks if it is controlled properly.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 covers uses, risks, controls and a judgement, but each briefly; Level 3 adds the factors that determine risk (type, half-life, dose units, route) and why risk-benefit analysis is difficult.	
	• Level 3 names the principles (ALARA, BATNEEC, inverse square law, CPA / CGM) rather than describing practice loosely.	
	• The judgement in Level 3 is supported with evidence (radon vs nuclear industry doses); Level 2's is an opinion.	
10.2	1 mark for the point below. **1** Consumers of the electricity may live in a different country / region from the uranium miners or the communities near waste stores who bear the risks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 295-304; AQA spec 3.4.3.2.14.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Attributes:** accuracy, precision, representative samples, reliability, anomalies, validity – hypothesis and null hypothesis – preliminary studies • **Sampling design:** random vs systematic location and avoiding bias – timing (diurnal, seasonal, long-term, weather) – sample size and number and the running mean – standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • **Statistics:** variability, standard deviation and error bars, significance and p-values, correlation vs causation – choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • **Application to Paper 1 topics:** monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	

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